

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399858
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Hsieh; Yuan-Ming et al.

Method and device for reducing latency in a peripheral component interconnect express link

Abstract

A method for reducing latency in a peripheral component interconnect express (PCIe) link between a PCIe root complex and a PCIe device is provided. The method includes measuring, by the PCIe device, a throughput of data transmission that is sent to a buffer of the PCIe device. The method includes determining, by the PCIe device, a latency tolerance reporting (LTR) value based on the measured throughput. The method includes transmitting, by the PCIe device, the LTR value to the PCIe root complex at an estimated time point via the PCIe link, wherein the estimated time point is prior to a time at which a buffer switching operation occurs.

Inventors: Hsieh; Yuan-Ming (Hsinchu, TW), Chang; Wei-Cheng (Hsinchu, TW), Huang; Hsin-Hao (Hsinchu, TW)

Applicant: MEDIATEK INC. (Hsinchu, TW)

Family ID: 1000008778345

Assignee: MEDIATEK INC. (Hsinchu, TW)

Appl. No.: 18/527468

Filed: December 04, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250181544 A1	Jun. 05, 2025

Publication Classification

Int. Cl.: G06F13/42 (20060101)

U.S. Cl.:

Field of Classification Search

CPC: G06F (13/4221); G06F (2013/0023)

USPC: 710/52

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2016/0162421	12/2015	Xiong	710/52	G06F 13/20
2019/0138220	12/2018	Hahn	N/A	G06F 3/0611
2020/0358685	12/2019	Gupta	N/A	G06F 1/3206
2021/0067451	12/2020	Duenas	N/A	G06F 13/385
2021/0200298	12/2020	Branover	N/A	G06F 13/1668
2022/0200881	12/2021	Shusterman	N/A	H04W 24/08
2023/0118047	12/2022	Huang	714/47.2	G06F 11/2268
2024/0281049	12/2023	Aung	N/A	G06F 1/263

Primary Examiner: Shin; Christopher B

Attorney, Agent or Firm: McClure, Qualey & Rodack, LLP

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

- (1) The present disclosure generally relates to power management in computer systems. More specifically, aspects of the present disclosure relate to techniques for reducing latency in a peripheral component interconnect express (PCIe) link between a PCIe root complex and a PCIe device.
- Description of the Related Art
- (2) Many modern computer systems use a peripheral component interconnect express (PCIe) link to communicate between a PCIe root complex and a PCIe device. When a PCIe link is unused for a long period of time, the PCIe device may try to save power by putting the PCIe link into a power-saving mode. Typically, the more power that is saved by using the power-saving mode, the longer the amount of time it takes for the PCIe link to exit the power-saving mode and become operational again.
- (3) In a modem PCIe device, the memory type that buffers network data is selected according to the network conditions. External memory (for example, DRAM), which is larger and has longer latency, is selected when the modem PCIe device receives non-latency-awareness and low throughput of network data. External memory provides longer buffering time which allows the PCIe link to stay inactive in a power-saving mode for a longer time. Internal memory (for example, SRAM), which is smaller and has shorter latency, is selected when the modem PCIe device receives latency-awareness or high throughput of network data. Internal memory provides lower

latency and a higher data processing rate of the modem PCIe device, whereas it makes the PCIe link wake up more frequently and consume more power.

(4) The PCIe standard includes Latency Tolerance Reporting (LTR) as a mechanism for periodically notifying the PCIe root complex of the return time. The PCIe root complex may be able to make decisions about entering different power states, which have different wake-up times, based on the LTR value sent by the modem PCIe device.

(5) In common designs, the LTR value is updated just before the network conditions change. For example, when the modem PCIe device detects that the throughput of network data has become higher, the modem PCIe device may switch its buffer from the internal memory to the external memory. At the same time, the modem PCIe device updates from a higher LTR value to a lower LTR value to prevent the buffer in the modem PCIe device from filling up.

(6) However, since the LTR value is sent to the PCIe root complex after the buffer switching operation occurs, the PCIe root complex cannot update its configuration immediately, resulting in unexpected or unacceptable behavior and possibly a poor user experience.

(7) Therefore, a method and a device for reducing latency in a peripheral component interconnect express (PCIe) link between a PCIe root complex and a PCIe device to improve upon the above shortcomings.

BRIEF SUMMARY OF THE INVENTION

(8) The following summary is illustrative only and is not intended to be limiting in any way. That is, the following summary is provided to introduce concepts, highlights, benefits and advantages of the novel and non-obvious techniques described herein. Select, not all, implementations are described further in the detailed description below. Thus, the following summary is not intended to identify essential features of the claimed subject matter, nor is it intended for use in determining the scope of the claimed subject matter.

(9) Therefore, the main purpose of the present disclosure is to provide a method and a device for reducing latency in a peripheral component interconnect express (PCIe) link between a PCIe root complex and a PCIe device to enable the PCIe root complex to enter the suitable power mode earlier.

(10) In an exemplary embodiment, a method for reducing latency in a peripheral component interconnect express (PCIe) link between a PCIe root complex and a PCIe device is provided. The method comprises measuring, by the PCIe device, a throughput of data transmission that is sent to a buffer of the PCIe device. The method comprises determining, by the PCIe device, a latency tolerance reporting (LTR) value based on the measured throughput. The method comprises transmitting, by the PCIe device, the LTR value to the PCIe root complex at an estimated time point via the PCIe link, wherein the estimated time point is prior to a time at which a buffer switching operation occurs.

(11) In some embodiments, the estimated time point is determined based on the measured throughput.

(12) In some embodiments, the estimated time point is one millisecond or 500 microseconds before the time when the buffer switching operation occurs.

(13) In some embodiments, the PCIe device is connected to the PCIe root complex using a Peripheral Component Interconnect Express (PCIe) bus.

(14) In an exemplary embodiment, a device for reducing latency in a peripheral component interconnect express (PCIe) link is provided. The device comprises one or more processors; and one or more computer storage media for storing one or more computer-readable instructions. The processor is configured to drive the computer storage media to execute the following tasks. The processor measures a throughput of data transmission that is sent to a buffer of the device. The processor determines a latency tolerance reporting (LTR) value based on the measured throughput. The processor transmits the LTR value to a PCIe root complex at an estimated time point via the PCIe link, wherein the estimated time point is prior to a time at which a buffer switching operation

occurs.

(15) In an exemplary embodiment, a non-transitory computer-readable storage medium storing instructions that, when executed by a processor in a PCIe device, cause the device to perform a method for reducing latency in a peripheral component interconnect express (PCIe) link between a PCIe root complex and the PCIe device is provided. The method comprises measuring a throughput of data transmission that is sent to a buffer of the PCIe device. The method comprises determining a latency tolerance reporting (LTR) value based on the measured throughput. The method comprises transmitting the LTR value to the PCIe root complex at an estimated time point via the PCIe link, wherein the estimated time point is prior to a time at which a buffer switching operation occurs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the present disclosure, and are incorporated in and constitute a part of the present disclosure. The drawings illustrate implementations of the disclosure and, together with the description, serve to explain the principles of the disclosure. It should be appreciated that the drawings are not necessarily to scale as some components may be shown out of proportion to their size in actual implementation in order to clearly illustrate the concept of the present disclosure.

(2) FIG. 1 is a block diagram illustrating an example link environment operable to reduce latency according to an embodiment of the disclosure.

(3) FIG. 2 presents a block diagram illustrating the PCIe device in according to an embodiment of the present disclosure.

(4) FIG. 3 presents a block diagram illustrating an operating system with a driver operating on the PCIe root complex coupled over a PCIe link to the firmware operating on the PCIe device in accordance with the described embodiments.

(5) FIG. 4 is a flow chart illustrating a method for reducing latency in a peripheral component interconnect express (PCIe) link according to an embodiment of the disclosure.

(6) FIG. 5 illustrates an exemplary operating environment for implementing embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(7) Various aspects of the disclosure are described more fully below with reference to the accompanying drawings. This disclosure may, however, be embodied in many different forms and should not be construed as limited to any specific structure or function presented throughout this disclosure. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Based on the teachings herein one skilled in the art should appreciate that the scope of the disclosure is intended to cover any aspect of the disclosure disclosed herein, whether implemented independently of or combined with any other aspect of the disclosure. For example, an apparatus may be implemented or a method may be practiced using number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using another structure, functionality, or structure and functionality in addition to or other than the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

(8) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects. Furthermore, like numerals refer to like elements throughout the several views, and the articles “a” and “the” includes plural references, unless otherwise specified

in the description.

(9) It should be understood that when an element is referred to as being “connected” or “coupled” to another element, it may be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present. Other words used to describe the relationship between elements should be interpreted in a like fashion. (e.g., “between” versus “directly between”, “adjacent” versus “directly adjacent”, etc.).

(10) The described embodiments perform operations for reducing latency in a peripheral component interconnect express (PCIe) link between a PCIe root complex and a PCIe device. In the described embodiments, the PCIe link between the PCIe root complex and the PCIe device may conform to the PCI special interest group (PIC-SIG) PCIe specification (e.g., PCI Express 3.0 Base specification revision 3.0).

(11) FIG. 1 is a block diagram illustrating an example link environment **100** operable to reduce latency according to an embodiment of the disclosure. The link environment **100** at least comprises a peripheral component interconnect express (PCIe) root complex **110** and a PCIe device **120**, wherein the PCIe root complex **110** and the PCIe device **120** are connected by a PCIe bus **130**.

(12) The PCIe root complex **110** can be (or can be included in) any device that communicates with the PCIe device **120** over the PCIe bus **130**. For example, the PCIe root complex **110** can be (or can be included in) a laptop computer, desktop computer, a server, an appliance, a subnotebook/netbook, a tablet computer, a cellular phone, a personal digital assistant (PDA), a smartphone, or another device. Note that the PCIe root complex **110** can include other subsystems (not shown) including but not limited to communications subsystems, display subsystems, data collection subsystems, audio and/or video subsystems, alarm subsystems, media processing subsystems, input/output (I/O) subsystems, and/or one or more other processing subsystems (e.g., CPUs), or power subsystems (e.g., battery, battery management unit, and/or other power management subsystems).

(13) The PCIe device **120** is any device that can communicate with the PCIe root complex **110** over the PCIe bus **130**. The PCIe device **120** may be or may include one or more devices or systems (e.g., peripherals) that perform functionalities including, for example, wireless communications, storage, and/or graphics processing. Note that in some embodiments, the PCIe device **120** may be located inside the PCIe root complex **110**, while in some embodiments the PCIe device **120** may be located externally to the PCIe root complex **110**. The PCIe device **120** will be discussed in more detail below with respect to FIG. 2.

(14) A link **140** between the PCIe root complex **110** and the PCIe device **120** is a PCIe link that allows communication between the PCIe root complex **110** and the PCIe device **120** using a PCIe specification such as PCIe base specification 3.0 by PCI-SIG.

(15) The PCIe root complex **110** may at least comprise a system on a chip (SOC) **112** and a memory **114**. The SOC **112** at least comprises a processor **1122** and may comprise other subsystems (not shown) such as memory, counters, clocks, interface regulators, power management circuits, and/or analog and/or digital interfaces. Additionally, note that in some embodiments, more than one PCIe link **140** may be connected to the processor **1122**.

(16) The processor **1122** comprises one or more devices configured to perform computational operations. For example, the processor **1122** may comprise one or more microprocessors each with one or more cores, application-specific integrated circuits (ASICs), microcontrollers, and/or programmable-logic devices. In some embodiments, some or all of the functionality of the SOC **112** may be replaced by one or more circuits that may include one or more microprocessors and/or multicore processing systems.

(17) The memory **114** comprises one or more devices for storing data and/or instructions for subsystems in the PCIe root complex **110** including those on the SOC **112** such as the processor **1122**. The memory **114** may comprise dynamic random access memory (DRAM), static random

access memory (SRAM), and/or other types of memory, and may comprise mechanisms for controlling access to the memory. In some embodiments, the memory **114** comprises a memory hierarchy that comprises one or more caches coupled to a memory (not shown) in the SOC **112**. In some embodiments, the memory **114** is coupled to one or more high-capacity mass-storage devices (not shown). For example, the memory **114** may be coupled to a magnetic or optical drive, a solid-state drive, or another type of mass-storage device. In these embodiments, the memory **114** may be used by the PCIe root complex **110** as fast-access storage for often-used data, while the mass-storage device is used to store less frequently used data.

(18) Furthermore, in this context, the PCIe device **120** may be an endpoint as those terms are used in the PCIe standard.

(19) FIG. 2 presents a block diagram illustrating the PCIe device **120** in according to an embodiment of the present disclosure. The PCIe device **120** comprises a processing subsystem **210** and a memory subsystem **220** coupled to a bus **230**. The PCIe device **120** may comprise other subsystems (not shown) such as subsystems for wireless communications, mass storage, and/or graphics processing, in addition to subsystems (not shown) for coupling to and communicating over a PCIe bus **130**.

(20) The processing subsystem **210** is any processing subsystem configured to perform computational and/or logic operations that may be used in a PCIe device, and may be implemented in any technology, including but not limited to any type of hardware module, software, firmware, and/or any other general purpose or special purpose logic. The processing subsystem **210** may comprise but is not limited to one or more central processing units (CPUs), microprocessors, application-specific integrated circuit (ASIC) chips, field-programmable gate arrays (FPGAs), other programmable-logic devices, dedicated logic devices, and microcontrollers.

(21) The memory subsystem **220** comprises one or more devices for storing data and/or instructions for the processing subsystem **210** and other subsystems (not shown) in the PCIe device **120**. For example, the memory subsystem **220** may comprise a dynamic random access memory (DRAM), a static random access memory (SRAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM), a flash memory, and/or other types of memory. In addition, the memory subsystem **220** may comprise firmware and mechanisms for controlling access to a buffer, a memory or other subsystems (not shown) in the PCIe device **120**.

(22) The processing subsystem **210** and the memory subsystem **220** are coupled together using the bus **230**. The bus **230** is an electrical, optical, or electro-optical connection that these subsystems and, in some embodiments, other subsystems (not shown) in the PCIe device **120** may use to communicate commands and data among one another. Although only one bus **230** is shown for clarity, different embodiments may comprise a different number or configuration of electrical or other connections among the subsystems.

(23) Although the processing subsystem **210** and the memory subsystem **220** are shown as separate subsystems in FIG. 2, in some embodiments, one or both of these subsystems can be integrated into one or more other subsystems in the PCIe device **120**. Furthermore, in some embodiments, the PCIe device **120** may comprise one or more additional processing subsystems **210** and/or memory subsystems **220**, and although we use specific subsystems to describe the PCIe device **120**, in alternative embodiments, different subsystems may be present in the PCIe device **120**.

Additionally, although the processing subsystem **210** and the memory subsystem **220** are depicted as separate subsystems in FIG. 2, in some embodiments these and other subsystems (not shown) in the PCIe device **120** may be implemented on one integrated circuit.

(24) FIG. 3 presents a block diagram illustrating an operating system **310** with a driver **312** operating on the PCIe root complex **110** coupled over a PCIe link to the firmware **320** operating on the PCIe device **120** in accordance with the described embodiments.

(25) In some embodiments, the operating system **310** is stored (as program codes) in the memory **114** and executed by the processor **1122**. Generally, the operating system **310** serves as an

intermediary between a system hardware in the PCIe root complex **110** and applications executed by the processor **1122**, which can be, for example, an email application, a web browser, and a game application. The operating system **310** also comprises a driver **312** which enables the operating system **310** and other applications operating on the processor **1122** to communicate with the PCIe device **120**. The operating system **310** and its general functions are known in the art and hence are not described in detail.

(26) The PCIe device **120** comprises a firmware **320** which may be preloaded on the PCIe device **120** and/or dynamically loaded by the driver **312**, and generally includes data and/or programming used to operate and control the PCIe device **120**. The firmware **320** may be stored in the memory subsystem **220** in a read-only memory (ROM), a programmable read-only memory (PROM), and/or an erasable programmable read-only memory (EPROM), and executed on the processing subsystem **210**. In some embodiments, the firmware **320** may be partially or completely replaced by software stored in the memory subsystem **220** and operating on the processing subsystem **202**, and/or one or more hardware modules (not shown) in the PCIe device **120**.

(27) It should be understood that the PCIe root complex **110** and the PCIe device **120** shown in FIG. **1** is an example of one suitable link environment **100**. The PCIe root complex **110** and the PCIe device **120** shown in FIG. **1** can be implemented via any type of computing device, such as the computing device **500** described with reference to FIG. **5**, as shown in FIG. **5**.

(28) FIG. **4** is a flow chart **400** illustrating a method for reducing latency in a peripheral component interconnect express (PCIe) link according to an embodiment of the disclosure. In this embodiment, the method is applied to the processing subsystem **210** of the PCIe device **120** in FIG. **2**.

(29) In step **S405**, the PCIe device measures a throughput of data transmission that is sent to a buffer of the PCIe device.

(30) Then, in step **S410**, the PCIe device determines a latency tolerance reporting (LTR) value based on the measured throughput.

(31) In step **S415**, the PCIe device transmitting the LTR value to the PCIe root complex at an estimated time point via the PCIe link, wherein the estimated time point is prior to a time at which a buffer switching operation occurs.

(32) Specifically, after the PCIe device measure the throughput of the current data transmission that is sent to the buffer of the PCIe device, the PCIe device may determine the LTR value and dynamically adjust the estimated time point according to the throughput of the current data transmission. For example, in case of low throughput, the PCIe device increases the LTR value and delays the estimated time point to transmit the LTR value. In the case of high throughput, the PCIe device reduces the LTR value and updates the estimated time point to an earlier time when the PCIe device transmits the LTR value.

(33) In some embodiments, the estimated time point is one millisecond or 500 microseconds before the time when the buffer switching operation occurs and can be adjusted based on experimental results.

(34) In another embodiment, the PCIe root complex may select a suitable power state that comply with the LTR value from a plurality of power states after receiving the LTR value, and then receives the data transmitted from the PCIe device.

(35) As mentioned above, the method and device for reducing latency in a peripheral component interconnect express (PCIe) link in the disclosure may dynamically adjust the LTR value and the estimated time point when the LTR value is transmitted according to different cases of throughput of data transmission. Since the LTR value is transmitted before the buffer switching operation occurs, the PCIe root complex can switch the power states in advance according to the LTR value and receive the data transmitted from the PCIe device faster.

(36) Having described embodiments of the present disclosure, an exemplary operating environment in which embodiments of the present disclosure may be implemented is described below. Referring

to FIG. 5, an exemplary operating environment for implementing embodiments of the present disclosure is shown and generally known as a computing device **500**. The computing device **500** is merely an example of a suitable computing environment and is not intended to limit the scope of use or functionality of the disclosure. Neither should the computing device **500** be interpreted as having any dependency or requirement relating to any one or combination of components illustrated.

(37) The disclosure may be realized by means of the computer code or machine-useable instructions, including computer-executable instructions such as program modules, being executed by a computer or other machine, such as a personal data assistant (PDA) or other handheld device. Generally, program modules may include routines, programs, objects, components, data structures, etc., and refer to code that performs particular tasks or implements particular abstract data types. The disclosure may be implemented in a variety of system configurations, including hand-held devices, consumer electronics, general-purpose computers, more specialty computing devices, etc. The disclosure may also be implemented in distributed computing environments where tasks are performed by remote-processing devices that are linked by a communication network.

(38) With reference to FIG. 5, the computing device **500** may include a bus **510** that is directly or indirectly coupled to the following devices: one or more memories **512**, one or more processors **514**, one or more display components **516**, one or more input/output (I/O) ports **518**, one or more input/output components **520**, and an illustrative power supply **522**. The bus **510** may represent one or more kinds of busses (such as an address bus, data bus, or any combination thereof). Although the various blocks of FIG. 5 are shown with lines for the sake of clarity, and in reality, the boundaries of the various components are not specific. For example, the display component such as a display device may be considered an I/O component and the processor may include a memory.

(39) The computing device **500** typically includes a variety of computer-readable media. The computer-readable media can be any available media that can be accessed by computing device **500** and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, not limitation, computer-readable media may comprise computer storage media and communication media. The computer storage media may include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules or other data. The computer storage media may include, but not limit to, random access memory (RAM), read-only memory (ROM), electrically-erasable programmable read-only memory (EEPROM), flash memory or other memory technology, compact disc read-only memory (CD-ROM), digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the computing device **500**. The computer storage media may not comprise signals per se.

(40) The communication media typically embodies computer-readable instructions, data structures, program modules or other data in a modulated data signal such as a carrier wave or other transport mechanism and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, but not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media or any combination thereof.

(41) The memory **512** may include computer-storage media in the form of volatile and/or nonvolatile memory. The memory may be removable, non-removable, or a combination thereof. Exemplary hardware devices include solid-state memory, hard drives, optical-disc drives, etc. The computing device **500** includes one or more processors that read data from various entities such as the memory **512** or the I/O components **520**. The display component(s) **516** present data indications to a user or to another device. Exemplary presentation components include a display device,

speaker, printing component, vibrating component, etc.

(42) The I/O ports **518** allow the computing device **500** to be logically coupled to other devices including the I/O components **520**, some of which may be embedded. Illustrative components include a microphone, joystick, game pad, satellite dish, scanner, printer, wireless device, etc. The I/O components **520** may provide a natural user interface (NUI) that processes gestures, voice, or other physiological inputs generated by a user. For example, inputs may be transmitted to an appropriate network element for further processing. A NUI may be implemented to realize speech recognition, touch and stylus recognition, facial recognition, biometric recognition, gesture recognition both on screen and adjacent to the screen, air gestures, head and eye tracking, touch recognition associated with displays on the computing device **500**, or any combination thereof. The computing device **600** may be equipped with depth cameras, such as stereoscopic camera systems, infrared camera systems, RGB camera systems, or any combination thereof, to realize gesture detection and recognition. Furthermore, the computing device **500** may be equipped with accelerometers or gyroscopes that enable detection of motion. The output of the accelerometers or gyroscopes may be provided to the display of the computing device **500** to carry out immersive augmented reality or virtual reality.

(43) Furthermore, the processor **514** in the computing device **500** can execute the program code in the memory **512** to perform the above-described actions and steps or other descriptions herein.

(44) It should be understood that any specific order or hierarchy of steps in any disclosed process is an example of a sample approach. Based upon design preferences, it should be understood that the specific order or hierarchy of steps in the processes may be rearranged while remaining within the scope of the present disclosure. The accompanying method claims present elements of the various steps in a sample order, and are not meant to be limited to the specific order or hierarchy presented.

(45) Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having the same name (but for use of the ordinal term) to distinguish the claim elements.

(46) While the disclosure has been described by way of example and in terms of the preferred embodiments, it should be understood that the disclosure is not limited to the disclosed embodiments. On the contrary, it is intended to cover various modifications and similar arrangements (as would be apparent to those skilled in the art). Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

Claims

1. A method for reducing latency in a peripheral component interconnect express (PCIe) link between a PCIe root complex and a PCIe device, comprising: measuring, by the PCIe device, a throughput of data transmission that is sent to a buffer of the PCIe device; determining, by the PCIe device, a latency tolerance reporting (LTR) value based on the measured throughput; and transmitting, by the PCIe device, the LTR value to the PCIe root complex at an estimated time point via the PCIe link, wherein the estimated time point is prior to a time at which a buffer switching operation occurs.
2. The method for reducing latency in a peripheral component interconnect express (PCIe) link as claimed in claim 1, wherein the estimated time point is determined based on the measured throughput.
3. The method for reducing latency in a peripheral component interconnect express (PCIe) link as claimed in claim 1, wherein the estimated time point is one millisecond or 500 microseconds before the time when the buffer switching operation occurs.

4. The method for reducing latency in a peripheral component interconnect express (PCIe) link as claimed in claim 1, wherein the PCIe device is connected to the PCIe root complex using a Peripheral Component Interconnect Express (PCIe) bus.
 5. A device for reducing latency in a peripheral component interconnect express (PCIe) link, comprising: one or more processors; and one or more computer storage media for storing one or more computer-readable instructions, wherein the processor is configured to drive the computer storage media to execute the following tasks: measuring a throughput of data transmission that is sent to a buffer of the device; determining a latency tolerance reporting (LTR) value based on the measured throughput; and transmitting the LTR value to a PCIe root complex at an estimated time point via the PCIe link, wherein the estimated time point is prior to a time at which a buffer switching operation occurs.
 6. The device for reducing latency in a peripheral component interconnect express (PCIe) link as claimed in claim 5, wherein the estimated time point is determined based on the measured throughput.
 7. The device for reducing latency in a peripheral component interconnect express (PCIe) link as claimed in claim 5, wherein the estimated time point is one millisecond or 500 microseconds before the time when the buffer switching operation occurs.
 8. The device for reducing latency in a peripheral component interconnect express (PCIe) link as claimed in claim 5, wherein the PCIe device is connected to the PCIe root complex using a Peripheral Component Interconnect Express (PCIe) bus.
 9. The device for reducing latency in a peripheral component interconnect express (PCIe) link as claimed in claim 5, wherein the device is a PCIe device.
 10. A non-transitory computer-readable storage medium storing instructions that, when executed by a processor in a PCIe device, cause the device to perform a method for reducing latency in a peripheral component interconnect express (PCIe) link between a PCIe root complex and the PCIe device, the method comprising: measuring a throughput of data transmission that is sent to a buffer of the PCIe device; determining a latency tolerance reporting (LTR) value based on the measured throughput; and transmitting the LTR value to the PCIe root complex at an estimated time point via the PCIe link, wherein the estimated time point is prior to a time at which a buffer switching operation occurs.
 11. The non-transitory computer-readable storage medium as claimed in claim 10, wherein the estimated time point is determined based on the measured throughput.
 12. The non-transitory computer-readable storage medium as claimed in claim 10, wherein the estimated time point is one millisecond or 500 microseconds before the time when the buffer switching operation occurs.
 13. The non-transitory computer-readable storage medium as claimed in claim 10, wherein the PCIe device is connected to the PCIe root complex using a Peripheral Component Interconnect Express (PCIe) bus.
-